Power Supply Systems

Part II

Phase 1: Distribution Basics

Aim: To understand primary distribution, secondary distribution, feeder, distributor, service mains, radial system, ring main system and interconnected system from absolute zero.

Style: Simple English, SN ma'am style, diagram first, formula after understanding.

Target: Semester-question-ready theory and basic numerical foundation.

Subject:	Power Supply Systems
Section:	Part II
Phase:	1 – Distribution Basics
Level:	Absolute zero to semester-ready foundation

Academic article style notes

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1 Phase 1 Roadmap

Phase 1 Goal

In this phase, we learn the basic language of distribution system. After this phase, distributor numericals, ring-main problems, voltage-drop problems and substation questions will become easier.

Power distribution → Primary distribution → Secondary distribution → Feeder → Distributor → Service mains

Radial system → Ring-main system → Interconnected system → Comparison

Memory Hook

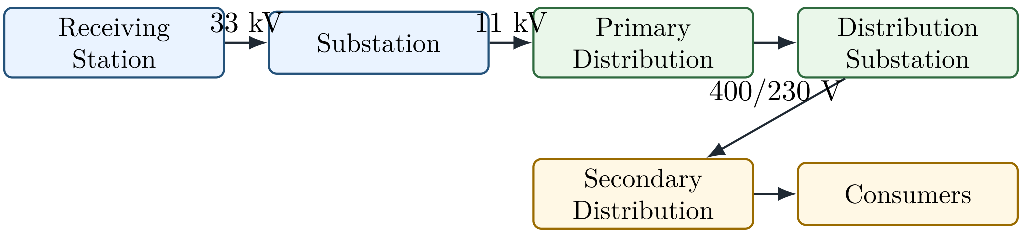
Phase 1 answers the question: **How does power go from substation to the consumer?**

2 What is Power Distribution?

Definition

Power distribution system is the part of the power supply system by which electrical power is supplied from substations to the consumers.

After transmission, the voltage is stepped down at receiving stations and substations. Then power is distributed to residential, commercial and industrial consumers.



Semester Answer Style

For a 4-mark answer, write:

1. Definition of distribution system.
2. Need of distribution system.
3. Diagram from substation to consumers.
4. Names of primary and secondary distribution.

3 Primary Distribution

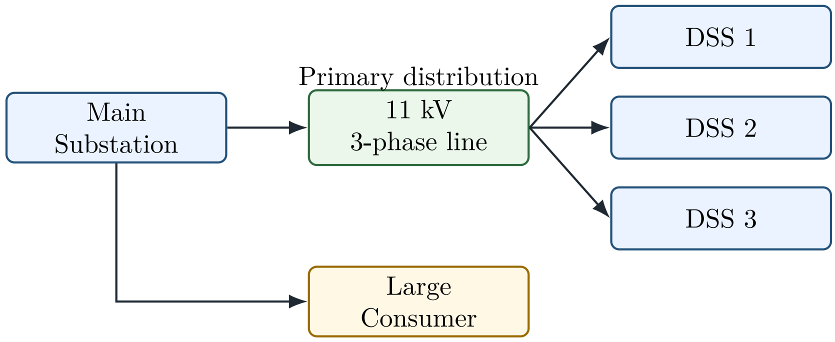
Definition

Primary distribution is the distribution of electrical power at medium voltage, generally around 11 kV, from substations to distribution substations or large consumers.

3.1 Important Points

- It is usually a 3-phase, 3-wire system.
- It supplies power from the main substation to distribution substations.
- Large consumers may also be supplied directly at this voltage.
- The voltage level is higher than consumer voltage, so current is lower and losses are reduced.

3.2 Primary Distribution Diagram



Memory Hook

Primary distribution means **power is still at high distribution voltage**. It is not yet the normal house voltage.

4 Secondary Distribution

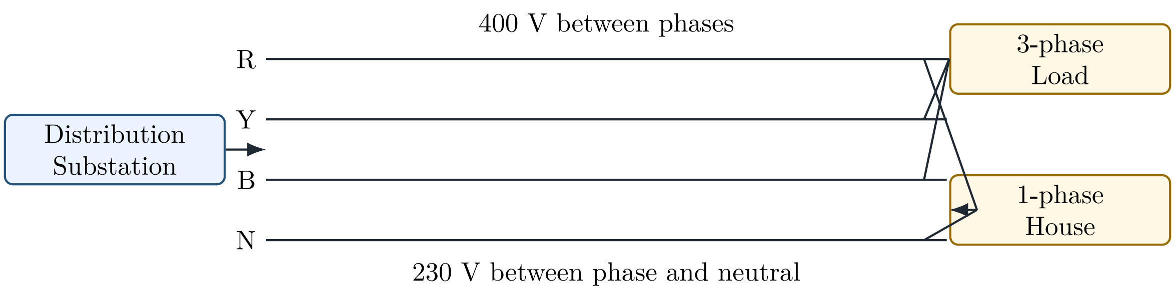
Definition

Secondary distribution is the distribution of electrical power at utilisation voltage, generally 400 V three-phase or 230 V single-phase, from distribution substations to consumers.

4.1 Important Points

- It is usually a 3-phase, 4-wire system.
- The fourth wire is neutral.
- Three-phase loads are connected across 400 V lines.
- Single-phase domestic loads are connected between phase and neutral, giving about 230 V.

4.2 Secondary Distribution Diagram



Semester Answer Style

If the question asks primary and secondary distribution, always write voltage level and wire system:

$$\boxed{\text{Primary: 11 kV, 3-phase, 3-wire}}$$

$$\boxed{\text{Secondary: 400/230 V, 3-phase, 4-wire}}$$

5 Difference Between Primary and Secondary Distribution

Point	**Primary Distribution**	**Secondary Distribution**
Voltage level	Higher distribution voltage, usually 11 kV	Consumer voltage, usually 400/230 V
Wire system	Usually 3-phase, 3-wire	Usually 3-phase, 4-wire
Purpose	Supplies distribution substations and large consumers	Supplies domestic, commercial and small industrial consumers
Current level	Lower current due to higher voltage	Higher current due to lower voltage
Voltage drop concern	Important but less severe	Very important because consumers are directly connected
Example	11 kV feeder from substation	400 V distributor in locality

6 Feeder, Distributor and Service Mains

6.1 Feeder

Definition

A **feeder** is a conductor which connects a substation to the area where power is to be distributed. Normally, no tapping is taken from a feeder.

$$\boxed{\text{Feeder is designed mainly for current carrying capacity.}}$$

6.2 Distributor

Definition

A **distributor** is a conductor from which several tappings are taken for supplying consumers.

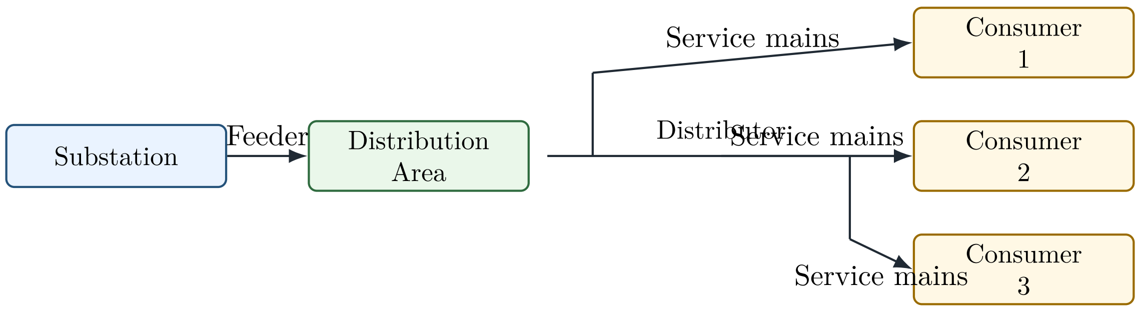
$$\boxed{\text{Distributor is designed mainly for permissible voltage drop.}}$$

6.3 Service Mains

Definition

Service mains are the conductors which connect the consumer's terminals to the distributor.

6.4 Diagram of Feeder, Distributor and Service Mains



6.5 Comparison Table

Point	Feeder	Distributor
Function	Carries power to distribution area	Supplies power to consumers through tappings
Tappings	Normally no tappings	Many tappings are taken
Design basis	Current carrying capacity	Voltage drop
Current	Almost same throughout, if no tapping	Decreases along the length as loads are tapped
Voltage drop	Less important than current rating	Very important
Example	Line from substation to locality	Line along street supplying houses

Memory Hook

Feeder = carries current. Distributor = controls voltage at consumers.

7 Why Feeder is Designed for Current Capacity

In a feeder, no consumer load is tapped along the length. So the same current generally flows throughout the feeder.

Therefore, the main question is:

Can the feeder safely carry this current without overheating?

So feeder design is based on:

- current carrying capacity,
- permissible temperature rise,
- mechanical strength,
- economy of conductor size.

Semester Answer Style

Semester line: A feeder is designed from the point of view of current carrying capacity because no tapping is normally taken from it and the current remains almost the same along its length.

8 Why Distributor is Designed for Voltage Drop

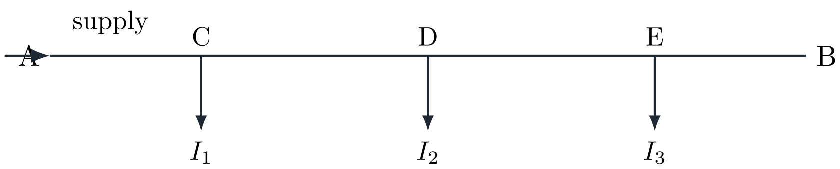
In a distributor, many consumers are connected at different points. As we move away from the feeding point, current changes section by section. So voltage also drops section by section.

The consumer voltage must remain within permissible limit.

$$\Delta V = IR$$

Therefore, distributor design is based on:

- voltage drop,
- current in each section,
- load position,
- permissible minimum voltage at consumer terminals.



Distributor with consumer tappings

Semester Answer Style

Semester line: A distributor is designed from the point of view of voltage drop because consumers are tapped at different points and the voltage at every consumer terminal must remain within the allowed limit.

9 Current Profile and Voltage Profile Along a Distributor

9.1 Meaning of Current Profile

Definition

Current profile means how current changes along different sections of a distributor.

For a distributor fed at one end, current is maximum near the feeding point and decreases as loads are tapped.

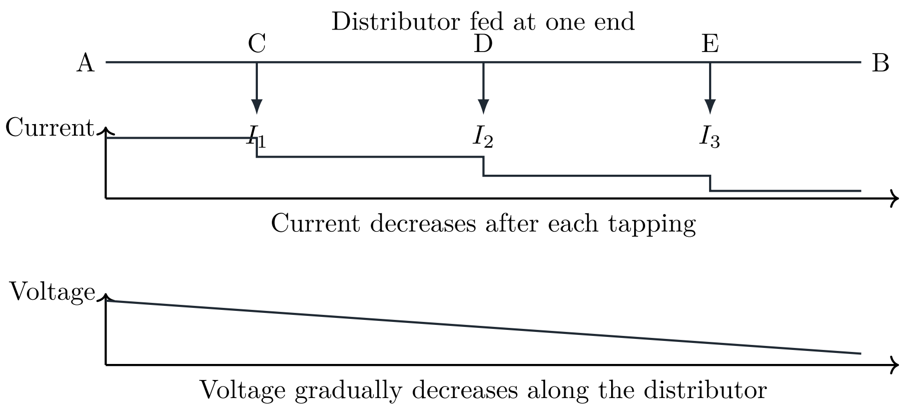
9.2 Meaning of Voltage Profile

Definition

Voltage profile means how voltage changes along the length of a distributor.

For a distributor fed at one end, voltage is maximum at the feeding point and minimum at the farthest end.

9.3 Simple Current and Voltage Profile Diagram



Common Mistake

Do not say current is the same in a distributor. Current changes because consumers are tapped at different points.

10 Types of Distribution Systems

The important distribution systems asked in semester are:

1. Radial distribution system
2. Ring-main distribution system
3. Interconnected distribution system

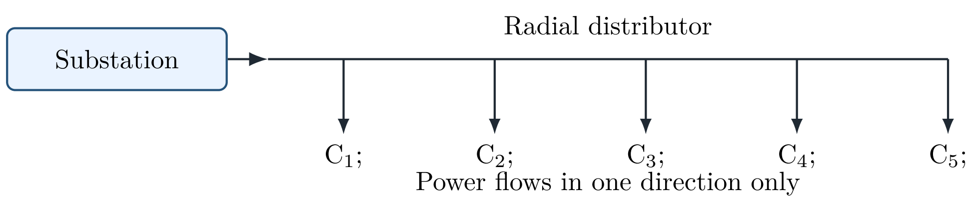
These are very important for theory questions.

11 Radial Distribution System

Definition

In a **radial distribution system**, the distributor is fed from one end only and power flows in one direction from the substation to consumers.

11.1 Diagram



11.2 Advantages

- Simple in construction.
- Low initial cost.
- Easy to understand and operate.

- Suitable for low-load rural or small distribution areas.

11.3 Disadvantages

- Reliability is poor.
- If fault occurs in one section, consumers beyond that point lose supply.
- Voltage regulation is poor at the far end.
- Voltage drop is large near farthest consumers.
- Not suitable for important urban loads.

Semester Answer Style

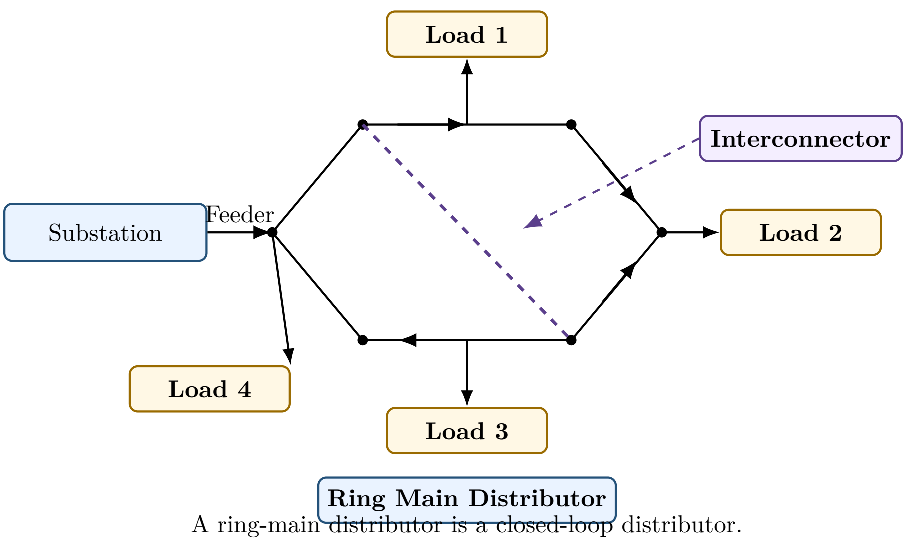
Most probable question: What are the drawbacks of radial distribution system?
Answer points:

1. Supply continuity is poor.
2. Fault in one section interrupts supply beyond fault point.
3. Voltage at far end is low.
4. More voltage drop occurs.
5. It is not suitable for dense and important load areas.

12 Ring-Main Distribution System

12.1 Diagram

_Diagram



A ring-main distributor is a closed-loop distributor. Loads are tapped from different points, and an interconnector improves reliability and voltage regulation.

12.2 Advantages

- More reliable than radial system.
- Supply can reach a consumer from two directions.

- If fault occurs in one section, the faulty section can be isolated.
- Continuity of supply is better.
- Voltage regulation is better than radial system.
- Copper saving may be possible due to better current distribution.

12.3 Disadvantages

- More complicated than radial system.
- Initial cost is higher.
- Protection and fault location are more difficult.
- Requires proper switching arrangement.

Memory Hook

Radial system is like a dead-end road. Ring main is like a circular road. If one road is blocked, supply can come from another side.

13 Ring Main with Interconnector

Definition

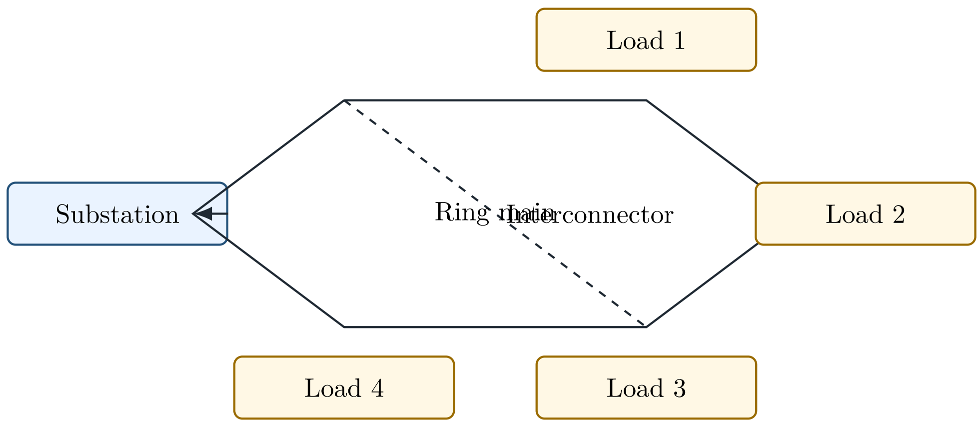
A **ring main with interconnector** is a ring-main system in which two points of the ring are connected by an additional conductor called interconnector.

13.1 Need of Interconnector

The interconnector improves:

- reliability,
- voltage regulation,
- flexibility of supply,
- load sharing between sections.

13.2 Diagram



Semester Answer Style

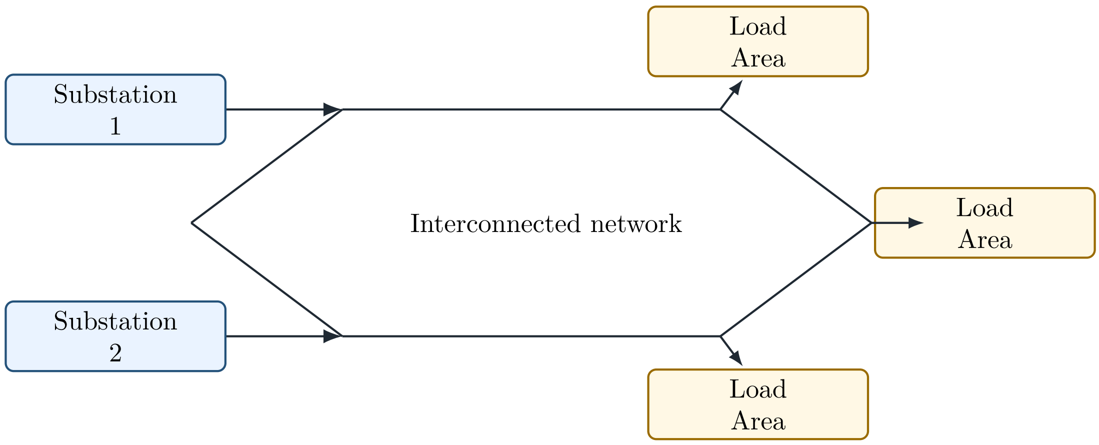
If a question asks "Differentiate radial system, ring-main system and ring-main with interconnector", draw all three small diagrams. This gives a strong semester impression.

14 Interconnected Distribution System

Definition

In an **interconnected distribution system**, two or more generating stations or substations feed the same distribution network.

14.1 Diagram



14.2 Advantages

- High reliability.
- Better continuity of supply.
- Load can be shared between substations.
- During fault or maintenance, supply may continue from another source.
- Better voltage regulation.

14.3 Disadvantages

- Cost is high.
- Protection system is complicated.
- Fault current may be high.
- Operation and coordination require more care.

15 Comparison of Radial, Ring-Main and Interconnected Systems

Point	**Radial System**	**Ring-Main System**	**Interconnected System**

Supply path	One path only	Two possible paths in loop	More than one source/path
Reliability	Lowest	Better	Highest
Cost	Lowest	Medium	Highest
Voltage regulation	Poor at far end	Better	Best
Fault effect	Consumers beyond fault lose supply	Faulty section can be isolated	Supply can be maintained from other source
Protection	Simple	Moderate	Complicated
Use	Rural/small areas	Urban distribution	Important and dense load areas

Semester Answer Style

For semester, the safest ranking is:

$$\boxed{\text{Reliability: Interconnected} > \text{Ring main} > \text{Radial}}$$

$$\boxed{\text{Cost: Radial} < \text{Ring main} < \text{Interconnected}}$$

16 Why Loop Arrangement is Better than Radial Arrangement

16.1 Meaning of Loop Arrangement

Loop arrangement means the distributor is connected in the form of a closed loop. A ring-main system is a loop arrangement.

16.2 Reasons

1. In radial system, supply comes from only one side. In loop system, supply can come from two sides.
2. If one section is faulty, that section can be isolated and other sections can remain alive.
3. Voltage drop is less because current is shared in two directions.
4. Voltage regulation is better.
5. Continuity of supply is improved.
6. It is more suitable for urban and important loads.

Semester Answer Style

Direct 4-mark answer: Loop arrangement is better than radial arrangement because it gives better reliability, better voltage regulation, lower voltage drop and continuity of supply during fault by isolating only the faulty section.

17 Comparison of Distribution Systems: AC and DC View

Although most modern distribution is AC, semester questions may ask general comparison.

17.1 DC Distribution

- DC distribution has no reactance drop.
- No skin effect occurs in DC.
- Voltage drop calculation is simpler.
- But DC voltage cannot be stepped up or stepped down by transformer.
- DC switching and interruption are more difficult.

17.2 AC Distribution

- AC voltage can be easily stepped up or stepped down by transformer.
- AC is suitable for general power distribution.
- But AC has reactance, power factor and skin effect problems.

Semester Answer Style

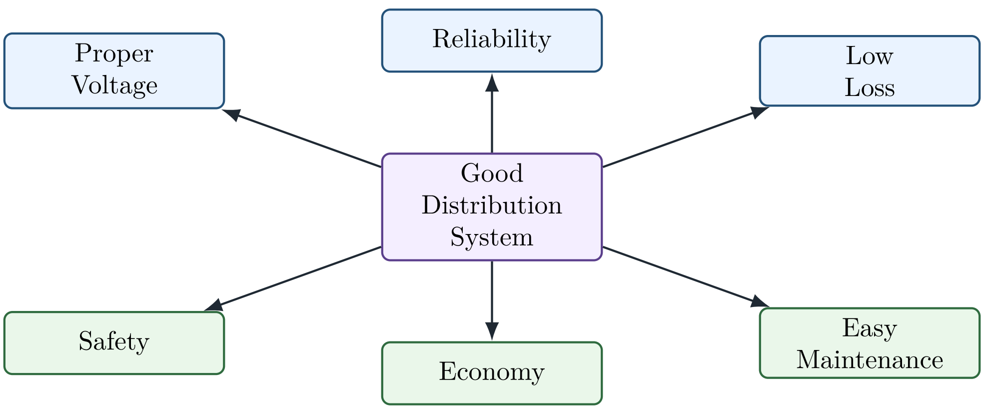
If asked in distribution context, write:

AC distribution is widely used because voltage transformation is easy by transformer.

18 Basic Design Requirements of a Distribution System

A good distribution system should have:

1. Proper voltage at consumer terminals.
2. Good reliability.
3. Low power loss.
4. Good safety.
5. Low cost.
6. Easy fault isolation.
7. Scope for future expansion.
8. Simple operation and maintenance.



19 Semester-Style Short Notes

19.1 Short Note: Primary Distribution

Primary distribution is the part of the distribution system in which electrical power is supplied at a higher distribution voltage, generally 11 kV. It connects main substations to distribution substations or large consumers. It is generally a 3-phase, 3-wire system. Since the voltage is high, current is reduced and line losses are comparatively less.

19.2 Short Note: Secondary Distribution

Secondary distribution is the part of the distribution system in which electrical power is supplied at utilisation voltage, generally 400 V three-phase or 230 V single-phase. It is generally a 3-phase, 4-wire system. The neutral wire is used to supply single-phase domestic consumers.

19.3 Short Note: Feeder

A feeder is a line conductor which connects the substation to the distribution area. Normally, no tapping is taken from it. Therefore, current remains nearly the same along its length. It is designed mainly on the basis of current carrying capacity.

19.4 Short Note: Distributor

A distributor is a conductor from which many tappings are taken to supply consumers. The current in the distributor changes from section to section. The voltage also drops along the length. Therefore, it is designed mainly from the point of view of voltage drop.

19.5 Short Note: Service Mains

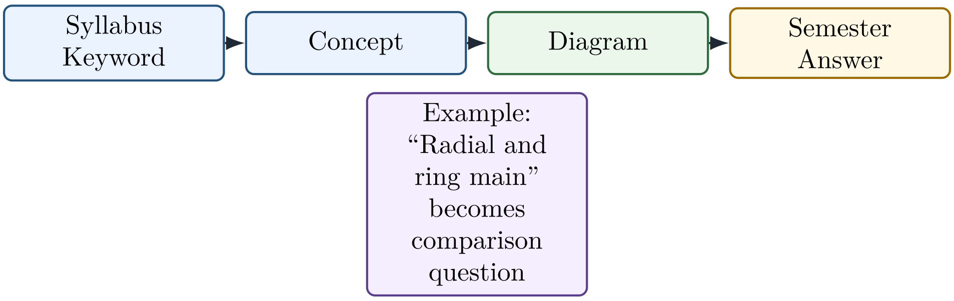
Service mains are the conductors which connect the consumer terminals to the distributor. They form the final connection between the distribution system and the consumer installation.

20 Most Predictable Semester Questions from Phase 1

No.	Question	Marks
1	What do you understand by feeder, distributor and service mains?	4–6
2	Explain why feeders are designed for current capacity and distributors for voltage drop.	4
3	Explain primary and secondary distribution with suitable diagram.	5–6
4	Differentiate between radial system, ring-main system and interconnected system.	6
5	Why is loop arrangement better than radial arrangement in distribution system?	4
6	What are the drawbacks of radial distribution system?	4–5
7	Explain ring-main distribution system with diagram.	5
8	Explain ring main with interconnector.	4–6

9	Mention the requirements of a good distribution system.	4
10	Draw current and voltage profile of a distributor fed at one end.	4

21 Reverse Engineering of Phase 1 PYQ Pattern



Syllabus Word	What It Means	Exam Question Form
Primary distribution	11 kV distribution from substation	Explain primary distribution.
Secondary distribution	400/230 V supply to consumers	Explain secondary distribution.
Feeder	Carries power, no tapping	Define feeder. Why current capacity design?
Distributor	Tappings to consumers	Define distributor. Why voltage drop design?
Radial system	One-side supply	Draw and discuss drawbacks.
Ring main	Closed loop supply	Explain advantages over radial.
Interconnector	Extra link in ring	Explain reliability improvement.

22 Phase 1 Exam Answer Templates

22.1 Template 1: Feeder, Distributor and Service Mains

Semester Answer Style

Answer format:

1. Draw substation–feeder–distributor–service mains diagram.
2. Define feeder.
3. Define distributor.
4. Define service mains.
5. Write feeder is current-capacity based.
6. Write distributor is voltage-drop based.

22.2 Template 2: Radial vs Ring Main

Semester Answer Style

Answer format:

1. Define radial system.
2. Draw radial system.
3. Define ring-main system.
4. Draw ring-main system.
5. Compare reliability, cost, voltage regulation and fault effect.
6. Conclude that ring main is better for important loads.

22.3 Template 3: Primary and Secondary Distribution

Semester Answer Style

Answer format:

1. Draw power path from substation to consumer.
2. Define primary distribution.
3. Mention 11 kV, 3-phase, 3-wire.
4. Define secondary distribution.
5. Mention 400/230 V, 3-phase, 4-wire.
6. Explain use of neutral wire.

23 Phase 1 Final Memory Sheet

One-Page Memory Sheet

1. Distribution system supplies power from substation to consumers.
2. Primary distribution is generally 11 kV, 3-phase, 3-wire.
3. Secondary distribution is generally 400/230 V, 3-phase, 4-wire.
4. Feeder connects substation to distribution area.
5. Distributor gives tappings to consumers.
6. Service mains connect consumer terminals to distributor.
7. Feeder is designed for current carrying capacity.
8. Distributor is designed for voltage drop.
9. Radial system is simple and cheap but less reliable.
10. Ring-main system is costlier but more reliable.
11. Interconnected system gives highest reliability but needs complex protection.
12. Loop arrangement is better because supply can come from two directions.

24 Phase 1 Self-Test

Answer these without seeing notes:

1. What is primary distribution?
2. What is secondary distribution?
3. Why is neutral wire used in secondary distribution?
4. What is feeder?
5. What is distributor?
6. What is service mains?
7. Why is feeder designed for current capacity?
8. Why is distributor designed for voltage drop?
9. Draw radial distribution system.
10. Draw ring-main distribution system.
11. Why is ring main better than radial system?
12. What is interconnected distribution system?

Common Mistake

If you cannot draw the diagrams from memory, do not move to distributor numericals yet. In Part-II, diagrams are the key to formulas.

End of Phase 1

Next Phase:
Phase 2 – Conductors and Comparison of Distribution Systems

Stranded conductor, ACSR, conductor materials, AC/DC distribution comparison, overhead/underground comparison, conductor-volume comparison foundation.